EE-102 Electric Circuit Analysis
Experiment # 3

To Investigate Characteristics of DC Series Circuit

Objective/s:

- To Investigate Characteristics of DC Series Circuit

Apparatus

- Digital Multimeter (DMM),
- Power Supply,
- Wire Stripper,
- Resistors: 100Ω, 3.6kΩ, 4.7kΩ, 6.8kΩ, 10kΩ

1. Series Circuit

In a series circuit, (Fig 3.1), the current is the same through all of the circuit elements.

The total Resistance $R_T = R_1 + R_2 + R_3$.

By Ohm's Law, the Current "I" is

$$I = \frac{E}{R_T}$$

Applying Kirchoff's Voltage Law around closed loop of Fig **3.**1, we find.

$$E = V_1 + V_2 + V_3$$

Where, $V_1 = IR_1, V_2 = IR_2, V_3 = IR_3$

Note in Fig 2.1, that I is the same throughout the Circuit.

The voltage divider rule states that the voltage across an element or across a series combination of elements in a series circuit is equal to the resistance of the element divided by total resistance of the series circuit and multiplied by the total applied voltage. For the elements of Fig 3.1

$$V_1 = \frac{R_1 E}{R_T}$$

$$V_2 = \frac{R_2 E}{R_T}$$

$$V_3 = \frac{R_3 E}{R_T}$$

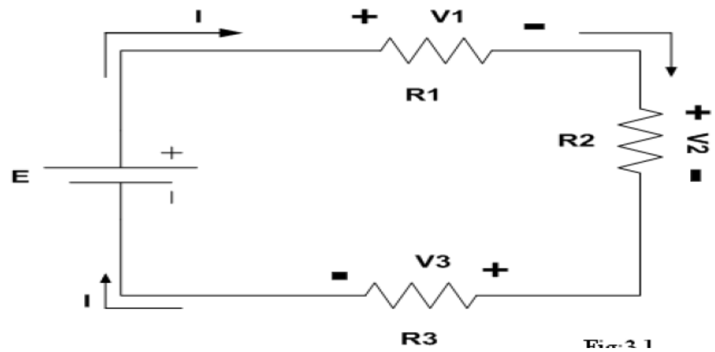

Figure 1

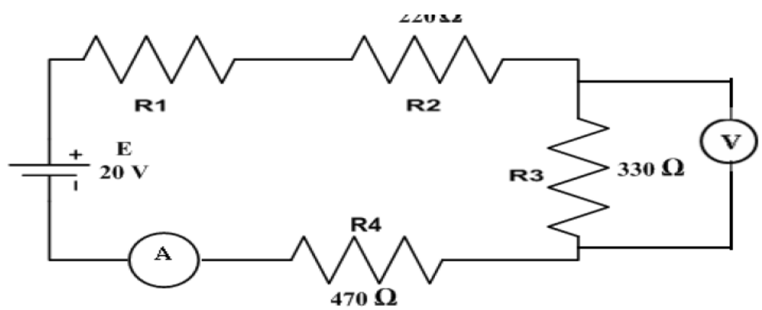

Figure 2

In a series circuit, (Fig 1a), the current is the same through all of the circuit elements.

The total Resistance $\mathbf{R_T = R_1 + R_2 + R_3}$.

By Ohm's Law, the Current "I" is

$$I = \frac{E}{R_T}$$

Applying Kirchoff's Voltage Law around closed loop of Fig 1a, We find.

$$E = V_1 + V_2 + V_3$$

Where, $V_1 = IR_1, V_2 = IR_2, V_3 = IR_3$

Note in Fig 1a, that I is the same throughout the Circuit.

The voltage divider rule states that the voltage across an element or across a series combination of elements in a series circuit is equal to the resistance of the element divided by total resistance of the series circuit and multiplied by the total voltage. For the elements of Fig 3.1

$$V_1 = \frac{R_1 E}{R_T}$$

$$V_2 = \frac{R_2 E}{R_T}$$

$$V_3 = \frac{R_3 E}{R_T}$$

2. Procedure

3. Construct the circuit shown in Fig 2.
4. Set the Dc supply to 20V by using DMM. Pick the resistances having values 220 ohm, 330 ohm& 430 ohm. Also verify their resistance by using DMM.
5. Measure voltage across each resistor with DMM and record it in the Table **2**.
6. Measure Current I delivered by source.
7. Shut down and disconnect the power supply. Then measure input resistance R_T across points A-E using DMM. Record that value.
8. Now Calculate, respective currents

Using

$$I_1 = \frac{V_1}{R_1}$$

and

$$R_T = \frac{E}{I}$$

9. Calculate V_1 & V_2 using voltage divider rule and measured resistance value.
10. Create an open circuit by removing R_3 & measure all voltages and current I.

3. Tasks

1. Resistors

S.No.	Normal Values (Ω)	Measured Values (Ω)	R_T (Measured) (Ω)	R_T (Calculated) (=E/I) (Ω)
1.	R_1 = 220			
2.	R_2 = 220			
3.	R_3 = 330			
4.	R_4 = 430			

Table 1

2. Voltage

S.No.	Measured Values (V)	Calculated Values (VDR)	Measured Values when R_3 is Open circuited (V)
1.			
2.			
3.			
4.			

Table 2

3. Current

S.No.	**Calculated Value (A) (Ohm's law)**	**Measured Value of I (A)**	**Measured I when R_3 is Open circuited (A)**
1.	**I_1=**		
2.	**I_2=**		
3.	**I_3=**		
4.	**I_4=**		

Table 3